

# Conformalized matrix completion and beyond

Yu Gui

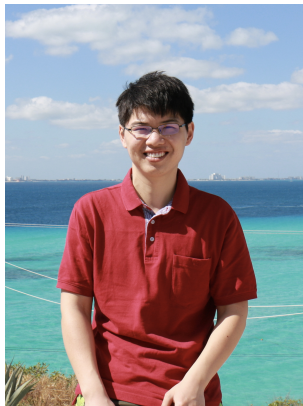
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Rina Foygel Barber



Cong Ma

# Matrix completion

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- ▶ Econometrics: panel data prediction and inference ([Amjad et al., 2018](#); [Athey et al., 2021](#))
- ▶ Recommender system: collaborative filtering ([Rennie and Srebro, 2005](#))
- ▶ Bioinformatics: gene-disease association ([Natarajan and Dhillon, 2014](#))
- ▶ ...

# Point completion

- Noisy and incomplete observation:

$$M_{ij} = M_{ij}^* + \text{noise} \quad (i, j) \in \Omega \subseteq [d_1] \times [d_2]$$

- Point estimation: estimate  $\mathbf{M}^*$  (with low complexity)
  - Convex optimization: Candes and Plan (2010); Negahban and Wainwright (2012); Koltchinskii et al. (2011); Chen et al. (2020); ...
  - nonconvex optimization: Keshavan et al. (2010); Ma et al. (2020); ...
- Point prediction: predict stochastic entries  $M_{ij}$  for  $(i, j) \in \Omega^c$  (Srebro and Shraibman, 2005; Foygel and Srebro, 2011); ...

# Success of model-based matrix completion

## ► Model assumptions

- Low-rank matrix:  $\text{rank}(\mathbf{M}^*) = r = O(1)$
- Random sampling:  $\mathbb{P}((i, j) \in \Omega) = p$  independently
- Random i.i.d. sub-Gaussian noise
- Incoherent and well-conditioned

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## Minimax rate

$$\|\widehat{\mathbf{M}} - \mathbf{M}^*\|_F \asymp \sigma \sqrt{n/p}$$

## ► Computational limit: [Chen et al. \(2020\)](#)

# From points to intervals

- *Confidence in the point completion?*
- *What if the model is wrong?*  
*How can we quantify the uncertainty in completed entries?*

## Goal

Construct a confidence interval  $C(i, j)$  for each  $(i, j) \in \Omega^c$  such that

$$\mathbb{P}(M_{ij} \in C(i, j)) \approx 1 - \alpha$$

# Model-based inference is feasible

- ▶ Asymptotic distribution (Chen et al., 2019)

$$\widehat{M}_{ij} - M_{ij}^* \approx \mathcal{N}(0, \theta_{ij}^2)$$

where  $\theta_{ij}^2 = (\sigma^2/p) (\|\mathbf{U}_i\|^2 + \|\mathbf{V}_j\|^2)$  and  $\mathbf{M}^* = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$

- ▶ Asymptotically valid  $(1 - \alpha)$ -confidence interval for  $M_{ij}^*$

$$C^*(i, j) = \widehat{M}_{ij} \pm q_{1-\alpha/2} \widehat{\theta}_{ij}$$

- ▶ Asymptotically valid  $(1 - \alpha)$ -confidence interval for  $M_{ij}$

$$C(i, j) = \widehat{M}_{ij} \pm q_{1-\alpha/2} \sqrt{\widehat{\theta}_{ij}^2 + \widehat{\sigma}^2}$$

Question: Is distribution-free inference possible?

# Distribution-free inference

- Free of model assumptions on the underlying matrix  $\mathbf{M}$
- Free of requirement of estimation algorithms
- Finite-sample guarantee?

# A detour: conformal prediction

- ▶ Any distribution  $P_{X,Y}$  (completely unknown)
- ▶  $\{(X_i, Y_i)\}_{i \leq n+1} \sim P_{X,Y}$  are exchangeable with unobserved  $Y_{n+1}$

## Finite-sample validity

Construct marginal confidence intervals any  $\alpha \in (0, 1)$

$$\mathbb{P}(Y_{n+1} \in C_{1-\alpha}(X_{n+1})) \geq 1 - \alpha$$

# Split conformal prediction

- ▶ Split dataset into a training set and a calibration set  $\mathcal{D}_{\text{cal}} = \{(X_i, Y_i)\}_{i \leq n}$
- ▶ Prefit  $\hat{\mu} : \mathcal{X} \rightarrow \mathcal{Y}$  on the training set  $\implies$  nonconformity score  $R(x, y)$

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- Prefit  $\hat{\mu} : \mathcal{X} \rightarrow \mathcal{Y}$  on the training set  $\implies$  nonconformity score  $R(x, y)$
- Exchangeability of  $\mathcal{D}_{\text{cal}} \cup \{(X_{n+1}, Y_{n+1})\}$

$$\left( R(X_{n+1}, Y_{n+1}) \mid \{R(x_i, y_i)\}_{i \leq n+1} \right) \sim \frac{1}{n+1} \sum_{i=1}^{n+1} \delta_{R(x_i, y_i)}$$

- Calculate the quantile

$$q_{1-\alpha} = \text{Quantile}_{1-\alpha} \left( \frac{1}{n+1} \sum_{i=1}^{n+1} \delta_{R(x_i, y_i)} + \frac{1}{n+1} \delta_{\infty} \right)$$

- Construct the prediction interval

$$C_{1-\alpha}(X_{n+1}) = \left\{ y : R(X_{n+1}, y) \leq q_{1-\alpha} \right\}$$

# Two ingredients of conformal prediction

- ▶ Exchangeable data  $\{(X_i, Y_i)\}_{i \leq n+1}$
- ▶ Symmetric algorithm  $\mathcal{A}$  (not required in split conformal prediction)

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**Question:** What if  $\{(X_i, Y_i)\}_{i \leq n+1}$  are not exchangeable? How can we fix this?

# CP under weighted exchangeability

## ► Weighted exchangeability

### Definition (Tibshirani et al., 2019)

Random variables  $\{V_i\}_{i \leq n+1}$  are said to be weighted exchangeable with weight functions  $\{w_i\}_{i \leq n+1}$  if the joint density can be factorized by

$$f(v_1, \dots, v_{n+1}) = \left\{ \prod_{i \leq n+1} w_i(v_i) \right\} \cdot g(v_1, \dots, v_{n+1})$$

where  $g$  is any function that does not depend on the ordering of its inputs.

# CP under weighted exchangeability

- If  $\{Z_i = (X_i, Y_i)\}_{i \leq n+1}$  are weighted exchangeable with weight functions  $w_i$

$$\left\{ R(Z_{n+1}) \middle| \{R(z_i)\}_{i \leq n+1} \right\} \sim \sum_{i \leq n+1} p_i(Z_1, \dots, Z_{n+1}) \delta_{R(Z_i)}$$

where  $p_i$ 's are standardized weights

$$p_i^w(z_1, \dots, z_{n+1}) = \frac{\sum_{\sigma: \sigma(n+1)=i} \prod_{j \leq n+1} w_j(z_{\sigma(j)})}{\sum_{\sigma} \prod_{j \leq n+1} w_j(z_{\sigma(j)})}, \quad i = 1, \dots, n+1$$

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- Construct the prediction interval

$$\hat{C}_{1-\alpha}(X_{n+1}) = \{y \in \mathcal{Y} : R(X_{n+1}, y) \leq q_{1-\alpha}^w\}$$

with the threshold

$$q_{1-\alpha}^w = \text{Quantile}_{1-\alpha} \left( \sum_{i \leq n} p_i^w \delta_{R(Z_i)} + p_{n+1}^w \delta_{\infty} \right)$$

# A special case: distribution shift

- Distribution shift

$$(X_i, Y_i) \sim P, \quad i \in [n] \quad (X_{n+1}, Y_{n+1}) \sim Q$$

- Joint density

$$f(z_1, \dots, z_{n+1}) = \frac{dQ}{dP}(z_{n+1}) \cdot \left( \prod_{i \leq n} dP(z_i) \right) \implies w_{n+1}(z) = \frac{dQ}{dP}(z)$$

- Standardized weights

$$p_i^w(z_1, \dots, z_{n+1}) = \frac{w_{n+1}(z_i)}{\sum_{j \leq n+1} w_{n+1}(z_j)} = \frac{\frac{dQ}{dP}(z_i)}{\sum_{j \leq n+1} \frac{dQ}{dP}(z_j)}, \quad i = 1, \dots, n+1$$

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Questions: Why is matrix completion special?

# Matrix completion under heterogeneous missingness

- **Heterogeneous sampling:** for a deterministic matrix  $\mathbf{M} \in \mathbb{R}^{d_1 \times d_2}$ , we assume

$$Z_{ij} = \mathbb{1}\{(i, j) \text{ is observed}\} \sim \text{Bern}(p_{ij}) \text{ independently}$$

for all  $(i, j) \in [d_1] \times [d_2]$  with  $p_{ij} \in (0, 1)$

$$\text{Set of observed indices: } \mathcal{S} = \{(i, j) : Z_{ij} = 1\}$$

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- **Questions:** Why is matrix completion special?

- ① How to apply data splitting to  $\mathcal{S}$  into a training set  $\mathcal{S}_{\text{tr}}$  and a calibration set  $\mathcal{S}_{\text{cal}}$ ? How to deal with the dependence between the two sets?
- ② How to address the setting where  $((i, j), M_{ij})$ 's are sampled without replacement?
- ③ For any  $(i_*, j_*) \in \mathcal{S}^c$ , does weighted exchangeability hold for  $\mathcal{S}_{\text{cal}} \cup \{(i_*, j_*)\}$  conditioning on  $\mathcal{S}_{\text{tr}}$ ?

# Data splitting

- For any pre-specified  $\eta \in (0, 1)$ , flip an unbalanced coin for each entry

$$W_{ij} \sim \text{Bern}(\eta).$$

- Split the observed set into  $\mathcal{S} = \mathcal{S}_{\text{tr}} \cup \mathcal{S}_{\text{cal}}$ :

$$\mathcal{S}_{\text{tr}} = \{(i, j) \in \mathcal{S} : W_{ij} = 1\}, \quad \mathcal{S}_{\text{cal}} = \{(i, j) \in \mathcal{S} : W_{ij} = 0\}.$$

$$\underbrace{\begin{bmatrix} \checkmark & 0 & 0 & 0 & \checkmark \\ 0 & 0 & \checkmark & \checkmark & 0 \\ \checkmark & 0 & 0 & \checkmark & 0 \\ 0 & 0 & \checkmark & 0 & 0 \\ \checkmark & 0 & 0 & \checkmark & 0 \end{bmatrix}}_{\mathbf{M}_{\mathcal{S}}} = \underbrace{\begin{bmatrix} \checkmark & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \checkmark & 0 \\ \checkmark & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ \checkmark & 0 & 0 & \checkmark & 0 \end{bmatrix}}_{\mathbf{M}_{\mathcal{S}_{\text{tr}}}} + \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 & \checkmark \\ 0 & 0 & \checkmark & 0 & 0 \\ 0 & 0 & 0 & \checkmark & 0 \\ 0 & 0 & \checkmark & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}}_{\mathbf{M}_{\mathcal{S}_{\text{cal}}}}$$

# Training-conditional weighted exchangeability

## Lemma

If  $(i_*, j_*) \mid \mathcal{S} \sim \text{Unif}(\mathcal{S}^c)$ , it holds that

$$\mathbb{P}\{(i_*, j_*) = (i_k, j_k) \mid \mathcal{S}_{\text{cal}} \cup \{(i_*, j_*)\} = \mathbb{I}^{(n_{\text{cal}}+1)}, \mathcal{S}_{\text{tr}}\} = w_{i_k j_k},$$

where  $\mathbb{I}^{(n_{\text{cal}}+1)} = \{(i_1, j_1), \dots, (i_{n_{\text{cal}}+1}, j_{n_{\text{cal}}+1})\}$  is the unordered set and

$$w_{i_k j_k} = \frac{h_{i_k j_k}}{\sum_{k'=1}^{n_{\text{cal}}+1} h_{i_{k'} j_{k'}}} \quad \text{with odds ratios } h_{ij} = \frac{1 - p_{ij}}{p_{ij}}$$

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$$h_{ij} = \frac{1 - p_{ij}}{p_{ij}} = \frac{dQ}{dP}(i, j)? \implies dP(i, j) = p_{ij}, \quad dQ(i, j) = 1 - p_{ij}$$

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$P$ : sampled subpopulation  $\xRightarrow{\text{shift}}$   $Q$ : unsampled subpopulation

# Conformalized matrix completion (cmc)

$$\underbrace{\begin{bmatrix} \checkmark & 0 & 0 & 0 & \checkmark \\ 0 & 0 & \checkmark & \checkmark & 0 \\ \checkmark & 0 & 0 & \checkmark & 0 \\ 0 & 0 & \checkmark & 0 & 0 \\ \checkmark & 0 & 0 & \checkmark & 0 \end{bmatrix}}_{\mathbf{M}_S} = \underbrace{\begin{bmatrix} \checkmark & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \checkmark & 0 \\ \checkmark & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ \checkmark & 0 & 0 & \checkmark & 0 \end{bmatrix}}_{\mathbf{M}_{S_{tr}}} + \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 & \checkmark \\ 0 & 0 & \checkmark & 0 & 0 \\ 0 & 0 & 0 & \checkmark & 0 \\ 0 & 0 & \checkmark & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}}_{\mathbf{M}_{S_{cal}}}$$

► With the training set  $\mathbf{M}_{S_{tr}}$

- 1 Estimate  $\widehat{\mathbf{M}} = (\widehat{M}_{ij})$  using any matrix completion algorithm
- 2 Estimate  $\widehat{\mathbf{P}} = (\widehat{p}_{ij})$  using any one-bit matrix completion algorithm
- 3 Calculate a local uncertainty estimate  $\widehat{\mathbf{s}} = (\widehat{s}_{ij})$

# Conformalized matrix completion (cmc)

$$\underbrace{\begin{bmatrix} \checkmark & 0 & 0 & 0 & \checkmark \\ 0 & 0 & \checkmark & \checkmark & 0 \\ \checkmark & 0 & 0 & \checkmark & 0 \\ 0 & 0 & \checkmark & 0 & 0 \\ \checkmark & 0 & 0 & \checkmark & 0 \end{bmatrix}}_{\mathbf{M}_{\mathcal{S}}} = \underbrace{\begin{bmatrix} \checkmark & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \checkmark & 0 \\ \checkmark & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ \checkmark & 0 & 0 & \checkmark & 0 \end{bmatrix}}_{\mathbf{M}_{\mathcal{S}_{\text{tr}}}} + \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 & \checkmark \\ 0 & 0 & \checkmark & 0 & 0 \\ 0 & 0 & 0 & \checkmark & 0 \\ 0 & 0 & \checkmark & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}}_{\mathbf{M}_{\mathcal{S}_{\text{cal}}}}$$

► With the calibration set  $\mathbf{M}_{\mathcal{S}_{\text{cal}}}$

- ① Calculate the nonconformity scores  $R_{ij} = \frac{|M_{ij} - \widehat{M}_{ij}|}{\widehat{s}_{ij}}, (i, j) \in \mathcal{S}_{\text{cal}}$
- ② Calculate the weights for each  $(i, j) \in \mathcal{S}_{\text{cal}} \cup \{(i_*, j_*)\}$

$$\widehat{w}_{ij} = \frac{\widehat{h}_{ij}}{\sum_{(i', j') \in \mathcal{S}_{\text{cal}}} \widehat{h}_{i'j'} + \widehat{h}_{i_*j_*}}$$

- ③ Output  $\widehat{C}(i_*, j_*) = \widehat{M}_{i_*j_*} \pm \widehat{q}_{i_*j_*} \widehat{s}_{i_*j_*}$  with

$$\widehat{q}_{i_*j_*} = \text{Quantile}_{1-\alpha} \left( \sum_{(i,j) \in \mathcal{S}_{\text{cal}}} \widehat{w}_{ij} \delta_{R_{ij}} + \widehat{w}_{i_*j_*} \delta_{\infty} \right)$$

- Define the average coverage rate over the unsampled set

$$\text{AvgCov}(\hat{C}; \mathbf{M}, \mathcal{S}) = \frac{1}{|\mathcal{S}^c|} \sum_{(i,j) \in \mathcal{S}^c} \mathbb{1}\{M_{ij} \in \hat{C}(i,j)\}$$

## Theorem

Conformalized matrix completion (cmc) satisfies

$$\mathbb{E}[\text{AvgCov}(\hat{C}; \mathbf{M}, \mathcal{S})] \geq 1 - \alpha - \mathbb{E}[\Delta].$$

- The coverage gap  $\mathbb{E}[\Delta]$  is defined with

$$\Delta = \frac{1}{2} \sum_{(i,j) \in \mathcal{S}_{\text{cal}} \cup \{(i_*, j_*)\}} |\hat{w}_{ij} - w_{ij}|$$

- ▶ Uniform sampling  $p_{ij} = p \implies \Delta = 0$

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- ▶ Logistic missingness

$$\log(p_{ij}/(1 - p_{ij})) = -\log(h_{ij}) = u_i + v_j, \quad \mathbf{u}^\top \mathbf{1} = 0$$

- 1 Consider the maximum likelihood estimate

$$\hat{\mathbf{u}}, \hat{\mathbf{v}} = \arg \max_{\mathbf{u}, \mathbf{v}} \mathcal{L}(\mathbf{u}, \mathbf{v}) \quad \text{subject to} \quad \mathbf{u}^\top \mathbf{1} = 0$$

- 2 Let  $\hat{p}_{ij} = (1 + \exp(-\hat{u}_i - \hat{v}_j))^{-1}$ , for which

$$\mathbb{E}[\Delta] \lesssim \sqrt{\frac{\log(\max\{d_1, d_2\})}{\min\{d_1, d_2\}}}$$

► Missingness with a general link function

$$\log(p_{ij}/(1 - p_{ij})) = -\log(h_{ij}) = \phi(A_{ij}), \quad \text{rank}(\mathbf{A}) = k^*, \quad \|\mathbf{A}\|_\infty \leq \tau$$

- ① Consider the maximum likelihood estimate

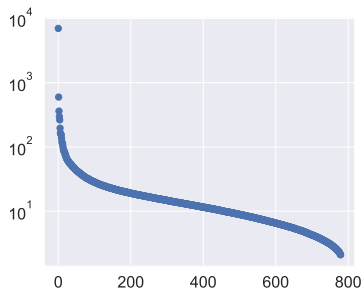
$$\hat{\mathbf{A}} = \arg \max_{\mathbf{B}} \mathcal{L}_{S_{\text{tr}}}(\mathbf{B}) \quad \text{subject to} \quad \|\mathbf{B}\|_* \leq \tau \sqrt{k^* d_1 d_2}, \quad \|\mathbf{B}\|_\infty \leq \tau$$

- ② Let  $\hat{h}_{ij} = \exp(-\phi(\hat{A}_{ij}))$ , for which

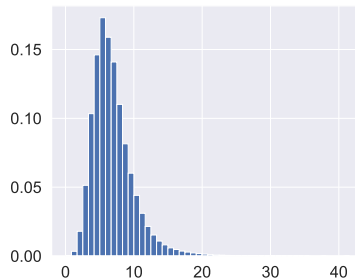
$$\mathbb{E}[\Delta] \lesssim \min\{d_1, d_2\}^{-1/4}$$

# Rossmann sales dataset<sup>1</sup>

- ▶ Daily sales (the unit is 1K dollar) of 1115 drug stores on workdays from Jan 1, 2013 to July 31, 2015



(a) Singular values of the underlying matrix  $M$ .



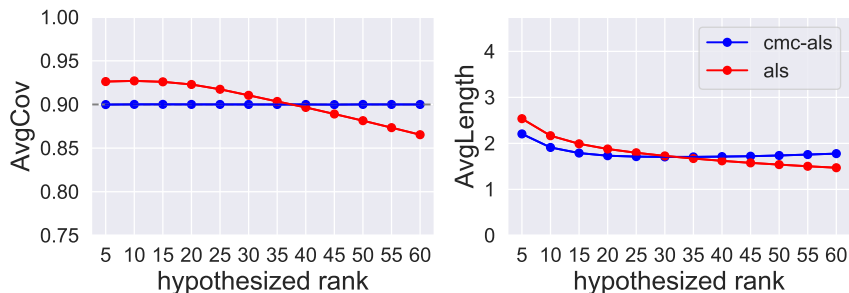
(b) Histogram of entries in  $M$ .

<sup>1</sup><https://www.kaggle.com/c/rossmann-store-sales>

# Well-specified missingness

## ► Settings of missingness

- 1 Homogeneous  $p_{ij}$ : apply random masking  $1 - Z_{ij} \sim \text{Bern}(0.2)$  independently



als vs cmc-als: homogeneous missingness.

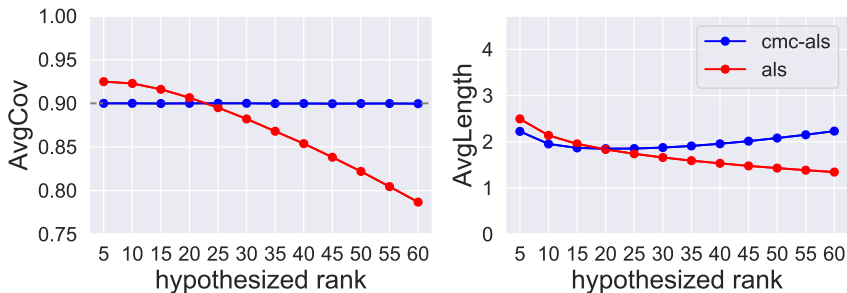
# Well-specified missingness

## ► Settings of missingness

- 1 Heterogeneous  $p_{ij}$ : Draw  $p_{ij}$ 's from

$$\log(p_{ij}/(1 - p_{ij})) = -\log(h_{ij}) = \phi(A_{ij})$$

with  $\text{rank}(\mathbf{A}) = 1$  and estimate  $\hat{\mathbf{P}}$  via the constrained maximum likelihood estimator



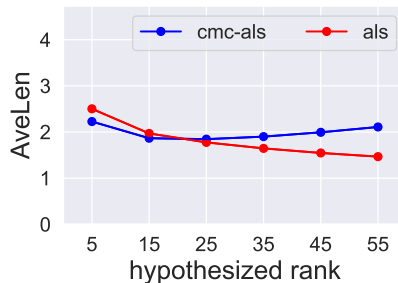
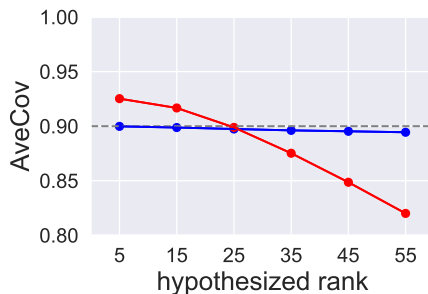
als vs cmc-als: logistic missingness.

# Misspecified missingness

## ► Settings of missingness

- 1 For weekdays, each entry is observed with  $p = 0.8$
- 2 On weekends, each entry is observed with a lower probability, e.g.  $0.8/3$
- 3 Randomly sample 200 stores, for which  $p = 0.8/3$

## ► Use one-bit matrix completion with logistic link to estimate $p_{ij}$



- ▶ Worst-case theoretical guarantee has a coverage gap

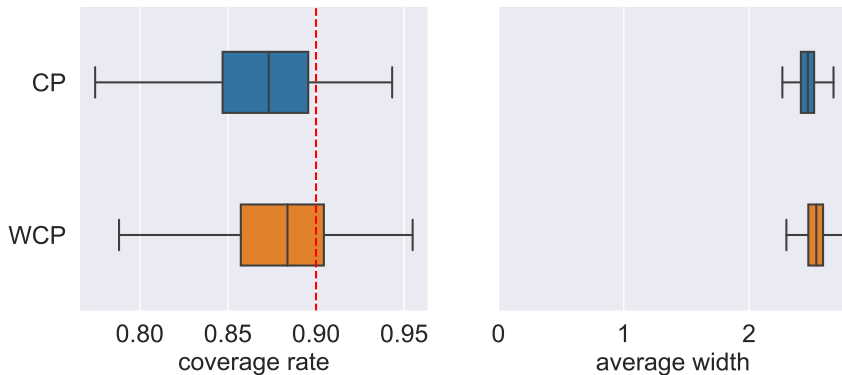
$$\mathbb{E}[\Delta] = \mathbb{E} \left[ \frac{1}{2} \sum_{(i,j) \in \mathcal{S}_{\text{cal}} \cup \{(i_*, j_*)\}} |\hat{w}_{ij} - w_{ij}| \right]$$

- ▶ When the sampling distribution is misspecified, a coverage gap is present in numerical experiments

*Question: Can we overcome the misspecification of density ratio  $w^*(x) = \frac{dP^{\text{test}}}{dP}(x)$  in reweighting methods?*

# An example in generic WCP

- Wine quality dataset<sup>2</sup>
- $\hat{w}$  is estimated via logistic regression with 2% of data from each group
- Estimated density ratio can be biased under misspecification



<sup>2</sup><https://archive.ics.uci.edu/dataset/186/wine+quality>

# A conservative approach

- Conditional miscoverage rate as risk

$$R_\alpha(X) = \mathbb{P}(Y \notin C_{1-\alpha}(X) \mid X)$$

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<sup>3</sup>Distributionally robust learning (DRL): Ben-Tal and Nemirovski (1998); El Ghaoui and Lebret (1997); Blanchet and Murthy (2019); Duchi and Namkoong (2018); etc

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- A conservative upper bound<sup>3</sup>

$$\mathbb{E}_{P^{\text{test}}}[R_\alpha(X)] \leq \sup_{Q \in \mathcal{Q}} \mathbb{E}_Q[R_\alpha(X)] \quad \text{as long as } P^{\text{test}} \in \mathcal{Q}$$

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- Confidence level calibration (Cauchois et al., 2020; Ai and Ren, 2024): choose  $\tilde{\alpha}$  such that

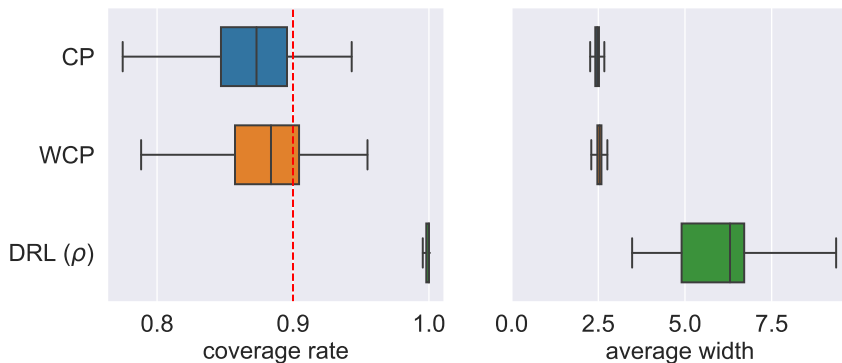
$$\sup_{Q \in \mathcal{Q}} \mathbb{E}_Q[R_{\tilde{\alpha}}(X)] \leq \alpha$$

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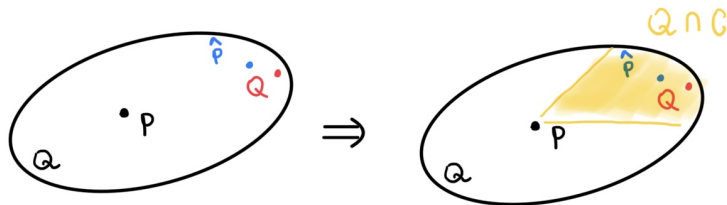
## Conservativeness of DR approach

$$\mathcal{Q} = \{Q : D_{\text{KL}}(Q||P) \leq \hat{\rho}\}, \quad \hat{\rho} \approx D_{\text{KL}}(P^{\text{test}}||P)$$



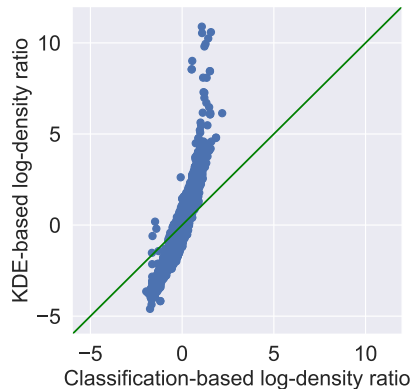
*Misspecification of reweighting methods VS Overly pessimism of DRL*

A middle ground?



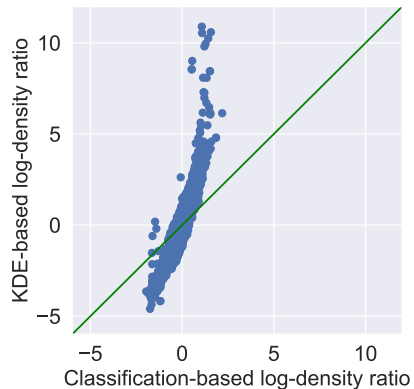
# A motivating example

**Fitted density ratio  $\hat{w}$  vs  $w^*$  proxy:** wine quality dataset



# A motivating example

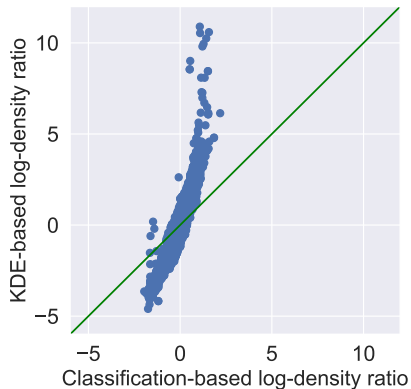
**Fitted density ratio  $\hat{w}$  vs  $w^*$  proxy:** wine quality dataset



- Approximately isotonic trend

# A motivating example

Fitted density ratio  $\hat{w}$  vs  $w^*$  proxy: wine quality dataset



- Approximately isotonic trend
- Under(Over)-represented regions in  $P^{\text{test}}$  are revealed by the under(over)-represented regions in  $\hat{P}$

- ▶ A fitted density ratio  $\hat{w}$  provides side information on the shape of  $w^*$

# DRL with shape constraints

- ▶ A fitted density ratio  $\hat{w}$  provides side information on the shape of  $w^*$
- ▶ Use the side information to construct an additional cone constraint

$$\mathcal{C}_{\hat{w}}^{\text{iso}} = \{\text{functions that are isotonic in } \hat{w}\}$$

# DRL with shape constraints

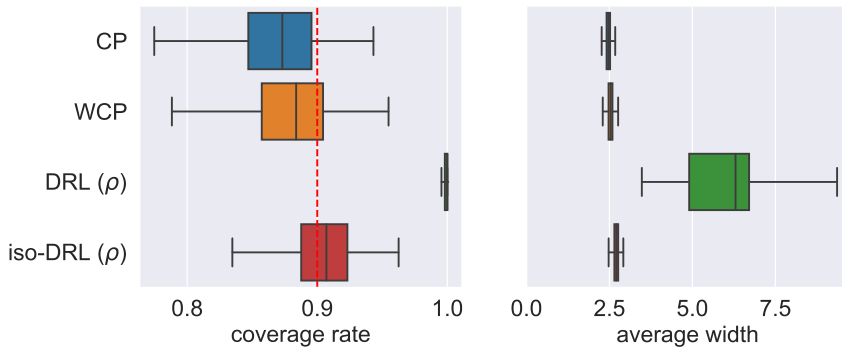
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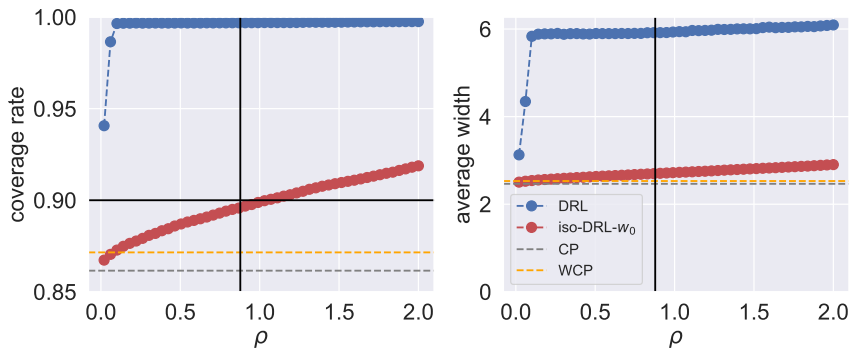
- ▶ Confidence level calibration via iso-DRL: choose  $\tilde{\alpha}$  such that

$$\sup_{\substack{Q \in \mathcal{Q} \\ dQ/dP \in \mathcal{C}_{\hat{w}}^{\text{iso}}}} \mathbb{E}_Q [R_{\tilde{\alpha}}(X)] \leq \alpha$$

# iso-DRL vs DRL



# Results with varying $\rho$



- ▶ Under any fixed partial order  $\preccurlyeq$  on  $\mathcal{X} \subseteq \mathbb{R}^d$ , denote the set of isotonic functions by  $\mathcal{C}^{\text{iso}}$  (closed and convex cone)

# iso-DRL under general partial orders

- ▶ Under any fixed partial order  $\preceq$  on  $\mathcal{X} \subseteq \mathbb{R}^d$ , denote the set of isotonic functions by  $\mathcal{C}^{\text{iso}}$  (closed and convex cone)
- ▶ iso-DRL focuses on the worst-case risk

$$\sup_{\substack{Q \in \mathcal{Q} \\ dQ/dP \in \mathcal{C}^{\text{iso}}}} \mathbb{E}_Q [R(X)]$$

# iso-DRL under general partial orders

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$$\sup_{\substack{Q \in \mathcal{Q} \\ dQ/dP \in \mathcal{C}^{\text{iso}}}} \mathbb{E}_Q [R(X)]$$

- ▶ Question:
  - Properties of population excess risk: how to solve the cone-constrained optimization problem in function space?
  - Consistent finite-sample estimate?

- Equivalent formulation without cone constraints

$$\sup_{\substack{Q \in \mathcal{Q} \\ dQ/dP \in \mathcal{C}^{\text{iso}}}} \mathbb{E}_Q [R(X)] = \sup_{Q \in \mathcal{Q}} \mathbb{E}_Q [R^{\text{iso}}(X)]$$

- Plug-in estimate with noisy risks for iso-DRL is consistent and inconsistent for DRL (Gui et al., 2024)
- iso-DRL is less sensitive to the choice of  $\mathcal{Q}$

# Open questions

- Extension to more general missingness: dependent sampling, block-wise missingness, ...
- Model-free confidence interval for the (effective) rank?
- Local coverage guarantee (conditioning on  $p_{i_*, j_*}$ )?
- ...

# Thank you!

## Papers

<https://neurips.cc/virtual/2023/poster/72747>

<https://arxiv.org/abs/2407.06867>

## Project pages

<https://github.com/yugjerry/conf-mc>

<https://github.com/yugjerry/iso-DRL>

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